

Tools Required

- Python 3.x
- Jupyter Notebook
- NumPy
- Pandas
- Matplotlib / Seaborn

Step 1: Sample Dataset (Housing Data)

	A	B	C	D	E
1	CrimeRate	Rooms	Distance	TaxRate	MedianPrice
2	6.43	3.13	4.03	311.61	39.46
3	6.8	7.46	1.96	410.96	11.19
4	2.26	5.53	1.29	299.42	36
5	5.49	4.1	7.48	604.72	10.26
6	8.08	6.49	4.74	277.74	48.29
7	3.43	3.46	2.06	623.75	34.15
8	8.09	6.65	6.9	686.56	25.14
9	5.57	7.15	7.8	630.85	33.09
10	7.08	3.23	3.51	344.69	13.19
11	2.4	3.51	4.06	517.84	24.59
12	3.76	4.05	3.94	668.33	35.92
13	6.13	3.86	9.02	281.7	25.18
14	9.9	6.2	7.13	542.31	43.71
15	7.78	4.15	1.35	357.73	20.71
16	2.19	7.71	10.64	357.34	36.22
17	4.02	7.57	6.05	332.44	19.87
18	5.66	4.31	7.43	648.91	25.98
19	2.27	7.99	6.6	245.45	11.88
20	1.19	6.14	9.71	411.08	12.54
21	3.88	7.98	6.82	685.54	44.43
22	0.21	6.6	8.5	468.49	20.67
23	6.45	3.56	5.78	426.86	48.15
24	8.77	4.32	6.51	289.33	46.51
25	8.72	4.49	8.03	504.49	16.11
26	7.65	5.7	9.56	465.18	10.02
27	3.31	3.1	11.22	639.36	43.27
28	3.14	3.29	10.66	673.47	13.43
29	4.91	3.35	9.37	582.92	15.14
30	4.81	5.75	3.92	636.22	26.93
31	2.2	5.7	9.03	300.58	22.47

Step 2: Import Libraries

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Step 3: Load Data

```
data = pd.read_excel("Housing data.xlsx")
data
```

Step 4: Descriptive Statistics

```
mean_values = data.mean()
variance_values = data.var()
```

```
mean_values
variance_values
```

Interpretation

- Mean → average value of each housing feature
- Variance → spread / variability in data

Step 5: Covariance Matrix

```
cov_matrix = data.cov()
cov_matrix
```

Interpretation

- Positive covariance → variables increase together
- Negative covariance → inverse relationship

Step 6: Correlation Matrix

```
corr_matrix = data.corr()
corr_matrix
```

Correlation ranges between **-1 and +1**

Step 7: Visualization

```
sns.heatmap(corr_matrix, annot=True, cmap="coolwarm")
plt.title("Correlation Matrix – Housing Features")
plt.show()
```

Step 8: Interpretation

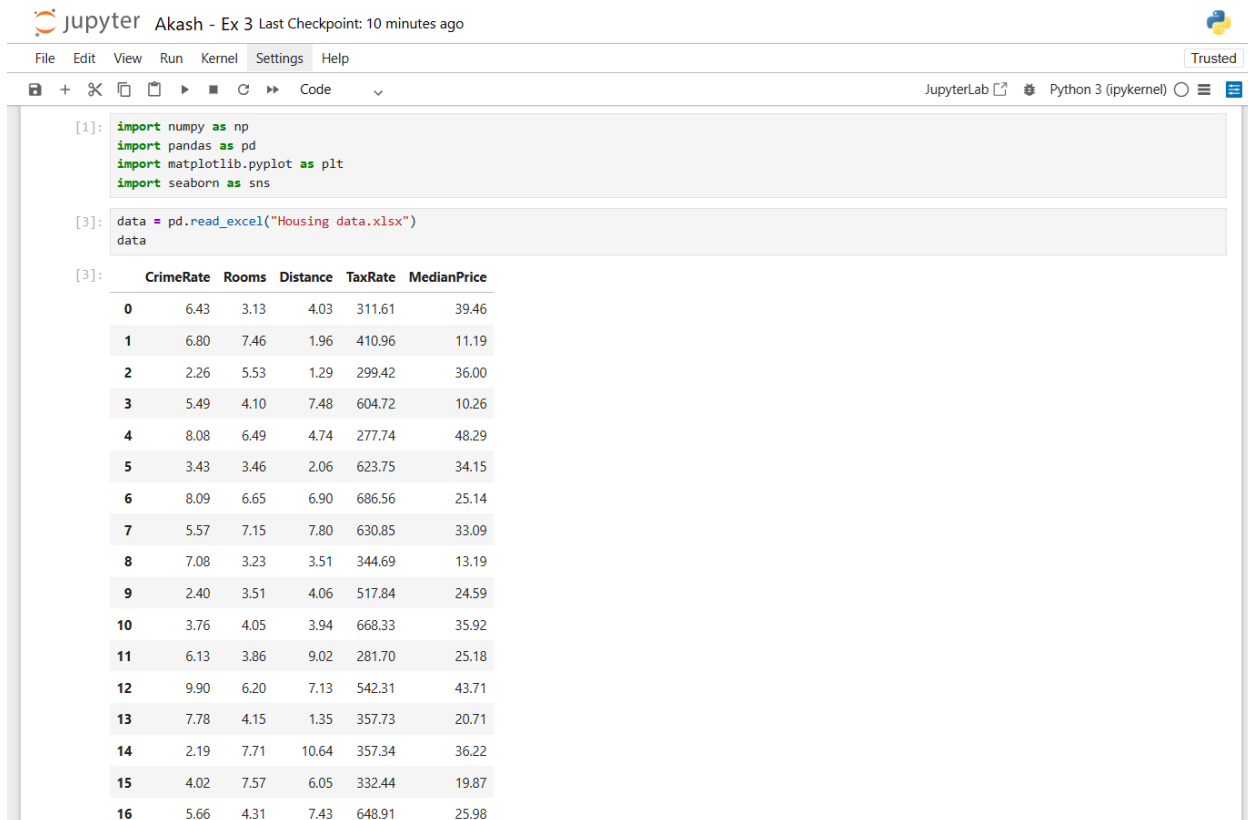
Relationships among features

- **Rooms vs MedianPrice** → Negative correlation
- **CrimeRate vs MedianPrice** → Positive correlation
- **Distance vs MedianPrice** → Moderate negative relationship
- **TaxRate vs MedianPrice** → Postive correlation

Inference

- Houses with more rooms have higher prices
- Higher crime reduces property value
- Homes closer to employment centers are costlier
- High tax rates reduce buyer demand and price

OUTPUT:



The image shows a JupyterLab interface with the following components:

- Header:** jupyter Akash - Ex 3 Last Checkpoint: 10 minutes ago
- Menu:** File Edit View Run Kernel Settings Help
- Toolbar:** Includes icons for file operations, running, and code execution.
- Code Editor:** Contains the following code:

```
[1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

[3]: data = pd.read_excel("Housing data.xlsx")
data
```
- Output:** A table with 17 rows and 6 columns. The first row is the header, and the subsequent rows are data points indexed from 0 to 16.

	CrimeRate	Rooms	Distance	TaxRate	MedianPrice
0	6.43	3.13	4.03	311.61	39.46
1	6.80	7.46	1.96	410.96	11.19
2	2.26	5.53	1.29	299.42	36.00
3	5.49	4.10	7.48	604.72	10.26
4	8.08	6.49	4.74	277.74	48.29
5	3.43	3.46	2.06	623.75	34.15
6	8.09	6.65	6.90	686.56	25.14
7	5.57	7.15	7.80	630.85	33.09
8	7.08	3.23	3.51	344.69	13.19
9	2.40	3.51	4.06	517.84	24.59
10	3.76	4.05	3.94	668.33	35.92
11	6.13	3.86	9.02	281.70	25.18
12	9.90	6.20	7.13	542.31	43.71
13	7.78	4.15	1.35	357.73	20.71
14	2.19	7.71	10.64	357.34	36.22
15	4.02	7.57	6.05	332.44	19.87
16	5.66	4.31	7.43	648.91	25.98

17	2.27	7.99	6.60	245.45	11.88
18	1.19	6.14	9.71	411.08	12.54
19	3.88	7.98	6.82	685.54	44.43
20	0.21	6.60	8.50	468.49	20.67
21	6.45	3.56	5.78	426.86	48.15
22	8.77	4.32	6.51	289.33	46.51
23	8.72	4.49	8.03	504.49	16.11
24	7.65	5.70	9.56	465.18	10.02
25	3.31	3.10	11.22	639.36	43.27
26	3.14	3.29	10.66	673.47	13.43
27	4.91	3.35	9.37	582.92	15.14
28	4.81	5.75	3.92	636.22	26.93
29	2.20	5.70	9.03	300.58	22.47

```
[4]: mean_values = data.mean()
     variance_values = data.var()

     mean_values
```

```
[4]: CrimeRate    5.086000
     Rooms        5.217667
     Distance     6.583333
     TaxRate      474.195667
     MedianPrice  27.150000
     dtype: float64
```

```
[5]: variance_values
```

```
[5]: variance_values
```

```
[5]: CrimeRate    6.452073
     Rooms        2.778370
     Distance     8.273092
     TaxRate      22694.535736
     MedianPrice  161.112745
     dtype: float64
```

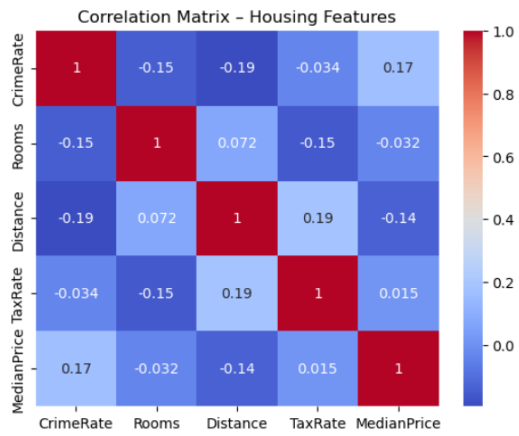
```
[6]: cov_matrix = data.cov()
     cov_matrix
```

	CrimeRate	Rooms	Distance	TaxRate	MedianPrice
CrimeRate	6.452073	-0.624954	-1.424331	-12.820897	5.337748
Rooms	-0.624954	2.778370	0.345518	-38.487697	-0.672359
Distance	-1.424331	0.345518	8.273092	80.221770	-4.996203
TaxRate	-12.820897	-38.487697	80.221770	22694.535736	28.556055
MedianPrice	5.337748	-0.672359	-4.996203	28.556055	161.112745

```
[7]: corr_matrix = data.corr()
     corr_matrix
```

	CrimeRate	Rooms	Distance	TaxRate	MedianPrice
CrimeRate	1.000000	-0.147606	-0.194952	-0.033505	0.165555
Rooms	-0.147606	1.000000	0.072068	-0.153273	-0.031779
Distance	-0.194952	0.072068	1.000000	0.185139	-0.136849
TaxRate	-0.033505	-0.153273	0.185139	1.000000	0.014934
MedianPrice	0.165555	-0.031779	-0.136849	0.014934	1.000000

```
[8]: sns.heatmap(corr_matrix, annot=True, cmap="coolwarm")  
plt.title("Correlation Matrix - Housing Features")  
plt.show()
```



RUBRICS:

Problem Understanding (10)	Design Document (10)	Implementation (15)	Output (5)	Time Management (5)	Viva (5)	Total (50)

RESULT:

Thus, housing feature data was successfully analyzed using descriptive statistics, covariance, and correlation analysis. Strong relationships between housing attributes and house prices were identified, aiding real estate valuation and prediction modeling decisions.